

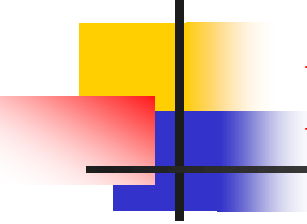
# **Digital Justice and Inclusion**

## **Digital Justice: Bridging the Technological Gap**

Dr. Ghaith Saker

2026

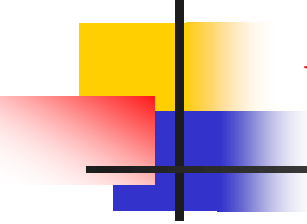
YPU



# Lecture Outline

---

1. Introduction to Digital Justice and Inclusion
2. Components of Digital Justice
3. The Digital Divide: Types and Causes
4. Impact of the Digital Divide on Societies
5. Strategies for Achieving Digital Inclusion
6. Case Studies and Practical Examples
7. Role of Governments and Institutions
8. Conclusion and Recommendations



# What is Digital Justice?

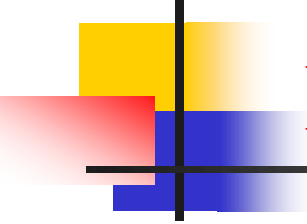
---

## **Digital Justice:**

A principle ensuring the fair distribution of digital technology benefits and opportunities while minimizing harms to all members of society without discrimination.

**Digital Inclusion:** Ensuring all individuals and groups can participate in the digital society, regardless of:

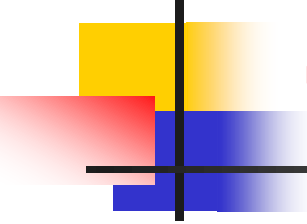
- Geographic location
- Economic status
- Age or gender
- Disability
- Digital skills



# Key Components of Digital Justice

---

1. **Physical Access:** Availability of devices and internet connectivity
2. **Affordability:** Reasonable cost of digital services
3. **Digital Skills:** Knowledge required for effective use
4. **Relevant Content:** Content available in multiple languages and levels
5. **Culture and Acceptance:** Societal acceptance and use of technology



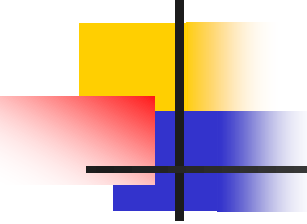
# The Digital Divide - Definition and Types

---

**Digital Divide :** The gap between individuals and regions in their ability to use information and communication technologies.

## Types:

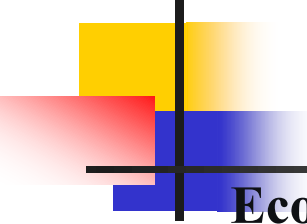
- **Access Divide:** Lack of infrastructure
- **Usage Divide:** Differences in usage skills
- **Quality Divide:** Disparities in service quality
- **Content Divide:** Lack of appropriate content for local languages and cultures



# Shocking Global Statistics

---

- **3.7 billion people** still lack internet access (World Bank, 2023)
- **In Africa:** 60% of population without internet vs. 14% in Europe
- **Gender Gap:** Women are 23% less connected in developing countries
- **Elderly:** 40% of people over 75 have never used the internet



# Causes of the Digital Divide

---

## **Economic:**

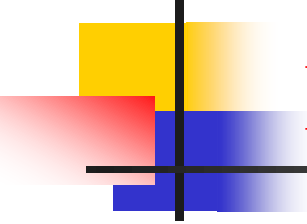
- High cost of devices and subscriptions
- Lack of purchasing power

## **Geographic:**

- Poor infrastructure in rural areas
- Difficulty delivering services to islands and remote regions

## **Social:**

- Digital illiteracy
- Language and cultural barriers
- Physical and sensory disabilities



# Impact of Digital Divide on Education

---

## **Before COVID-19:**

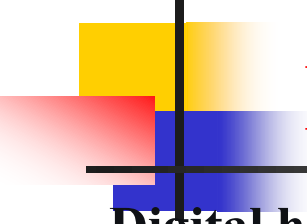
- 50% of students in developing countries without internet

## **During Pandemic:**

- **1.6 billion students** affected by school closures
- **500 million students** unable to access remote learning

**Practical Example:** In India, a student died by suicide because she couldn't afford a smartphone for online learning





# Impact on Healthcare

---

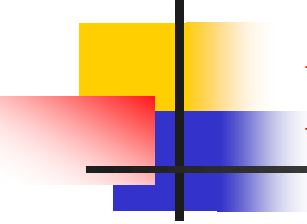
**Digital health has become essential:**

- Remote medical appointments
- Electronic health records
- Health applications

**Reality:**

- 40% of people in low-income countries cannot access digital health services
- Elderly and people with disabilities are most affected

**Example:** COVID-19 tracking apps weren't available in local languages in many African countries



# Impact on Economic Opportunities

---

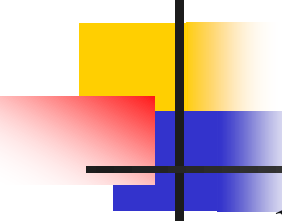
## **Job Market:**

- 80% of jobs require basic digital skills
- Unemployed people have lower digital literacy

## **Finance:**

- Digital banking services excluded the poor
- Small traders lost online selling opportunities

**Example:** In Kenya, M-PESA digital payment app helped include 80% of population in financial system



# Strategies for Digital Inclusion (1)

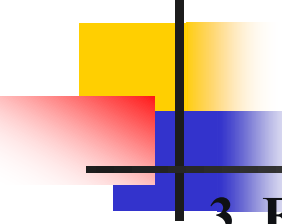
---

## **1. Infrastructure:**

- Expand broadband networks
- Establish community internet centers
- Public-private partnership projects

## **2. Affordability:**

- Provide affordable pricing for vulnerable groups
- Government device support programs
- Tax exemptions for basic digital services



# Strategies for Digital Inclusion (2)

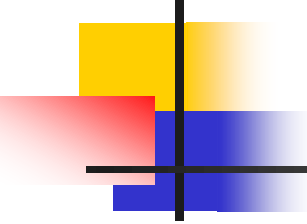
---

## **3. Education and Skills:**

- Integrate digital literacy into curricula
- Training programs for elderly
- Community training centers

## **4. Inclusive Design:**

- Accessible applications for people with disabilities
- Simple user interfaces
- Multilingual content

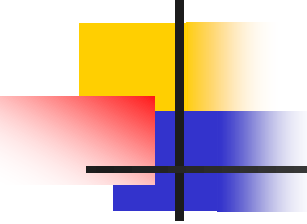


# Successful Case Studies (1)

---

## **Estonia: A Digital Republic**

- 99% of government services electronic
- Digital identity for every citizen
- "Digital Identity" program for foreign residents
- Results: Saves 2% of GDP annually

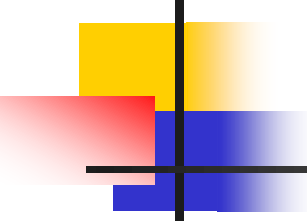


## Successful Case Studies (2)

---

### **Rwanda: "Digital Recovery"**

- 4G network covers 95% of population
- Free coding lessons for youth
- Medicine delivery by drones
- Results: Technology sector growing 15% annually

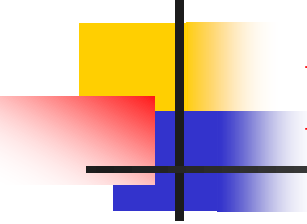


## Successful Case Studies (3)

---

### **India: "Biometric Identity"**

- Aadhaar program: Digital identity for 1.3 billion people
- Direct payment of social benefits
- Digital bank accounts for excluded populations
- Results: Saved \$23 billion by preventing corruption



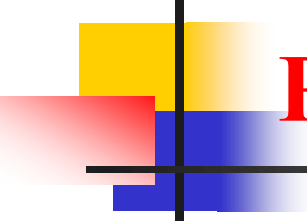
# Role of Governments and Policies

---

## Essential Policies:

- Recognize internet as basic service (like water and electricity)
- Laws prohibiting digital discrimination
- Mandatory standards for inclusive design
- Investments in rural and marginalized area infrastructure



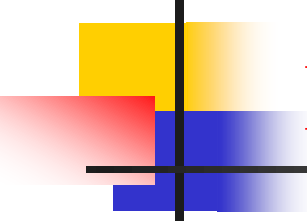


# Role of Private Sector and Companies

---

## **Corporate Responsibility:**

- Design inclusive and affordable products
- Social responsibility programs for marginalized groups
- Partnerships with governments and civil society
- Adopt inclusive design principles in development

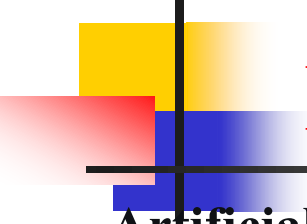


# Role of Individuals and Civil Society

---

## **What We Can Do:**

- Volunteer to train elderly in technology
- Advocate for fair digital policies
- Support community digital inclusion initiatives
- Raise awareness about importance of digital skills



# Future Challenges

---

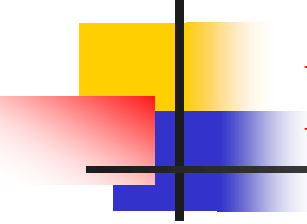
## **Artificial Intelligence:**

- Risk of amplifying existing biases
- Need for diverse and inclusive data

## **Emerging Technologies:**

- Virtual and augmented reality
- Internet of Things
- Quantum computing

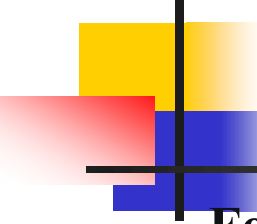
**Necessity:** Ensure these technologies are inclusive from design phase



# Digital Justice Measurement Indicators

---

1. Percentage of population with internet access
2. Average download and upload speeds
3. Monthly subscription cost as percentage of income
4. Percentage of individuals with basic digital skills
5. Percentage of local content on internet



# Recommendations

---

## **For Governments:**

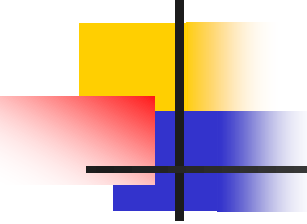
- Adopt national digital inclusion strategies
- Allocate special budgets for marginalized groups

## **For Educational Institutions:**

- Integrate digital skills at all levels
- Provide continuous teacher training

## **For Civil Society:**

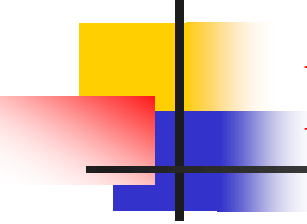
- Monitor and evaluate digital inclusion policies
- Document and share best practices



## Conclusion: Why Care About Digital Justice?

---

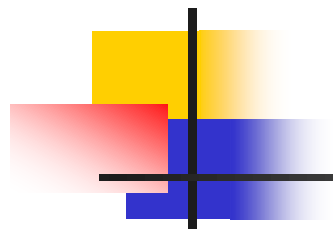
- **Ethical:** Right to access information is basic human right
- **Economic:** Every 10% increase in broadband penetration increases GDP by 1.3%
- **Social:** Reduce gaps and limit inequality
- **Developmental:** Achieve 2030 Sustainable Development Goals



# Discussion Questions

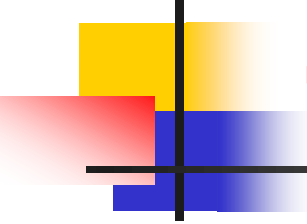
---

1. What are the biggest challenges to achieving digital justice in our local community?
2. How can emerging technologies like AI increase or decrease the digital divide?
3. What is the ethical responsibility of major tech companies in achieving digital inclusion?
4. How can we as individuals contribute to reducing the digital divide in our environment?



## **Review and Assessment**

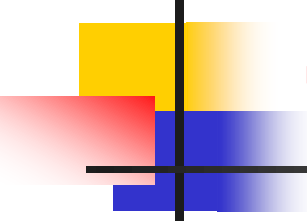




# True/False Questions

---

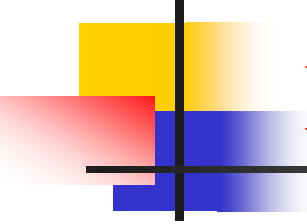
1. **Digital Justice focuses solely on providing affordable internet access to everyone. False.**
2. Digital Justice ensures the fair distribution of technology's benefits and opportunities for all, which includes access, affordability, skills, relevant content, and cultural acceptance. **True.**
3. **The "Quality Divide" refers to the lack of appropriate content in local languages. False.**
4. : The "Quality Divide" refers to disparities in service quality (like internet speed), while the "Content Divide" refers to the lack of appropriate content for local languages and cultures. **True.**
5. **The primary cause of the digital divide in rural areas is typically a lack of digital literacy. False.**
6. the primary cause is often geographic, meaning poor infrastructure and difficulty in delivering services, not just a lack of skills. **True.**



# True/False Questions

---

1. A key strategy for digital inclusion is to design complex user interfaces to ensure security. **False.**
2. A key strategy is "Inclusive Design," which involves creating simple user interfaces and accessible applications for people with disabilities. **True.**
3. Digital justice has a measurable economic benefit, such as increasing GDP with greater broadband penetration. **True.**
4. The "Usage Divide" describes the gap in the physical infrastructure for internet access. **False.**
5. The "Usage Divide" refers to differences in the skills needed to use the technology effectively. **True.**



# Multiple Choice Questions

---

1. Which of the following is NOT listed as a key component of Digital Justice?

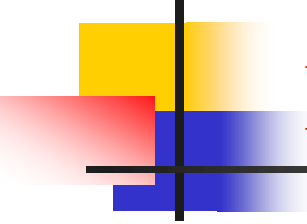
- a) Physical Access
- b) Affordability
- c) Digital Skills
- d) Maximum Internet Speed**

2. The "Digital Divide" is defined as the gap in what?

- a) The price of software between countries
- b) The ability to use information and communication technologies**
- c) The number of social media users between age groups
- d) The quality of computers available in schools

6. What is a primary strategy for improving the *affordability* of digital access?

- a) Building more satellites
- b) Government device support programs and tax exemptions**
- c) Integrating digital literacy into school curricula
- d) Developing more complex applications



# Multiple Choice Questions

---

**7. What is a recommended role for governments in promoting digital justice?**

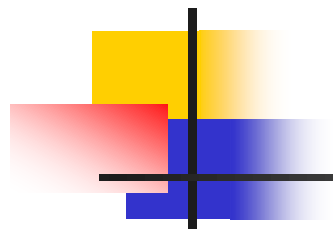
- a) Limiting internet access to reduce misinformation
- b) Recognizing internet access as a basic service, like water and electricity**
- c) Focusing investment only on urban technological hubs
- d) Reducing digital literacy programs to cut costs

**9. Which of the following is an indicator for measuring Digital Justice?**

- a) The number of tech companies in a country
- c) The monthly subscription cost as a percentage of income**
- b) The popularity of the latest social media app
- d) The number of programming languages taught in schools

**10. Why is Digital Justice important for economic reasons?**

- a) It guarantees that all websites will be free to access.
- b) Increased broadband penetration is directly linked to GDP growth.**
- c) It forces companies to lower the salaries of IT workers.
- d) It ensures that all jobs can be performed remotely.



**Thank You for Your Attention**

**Questions?**